

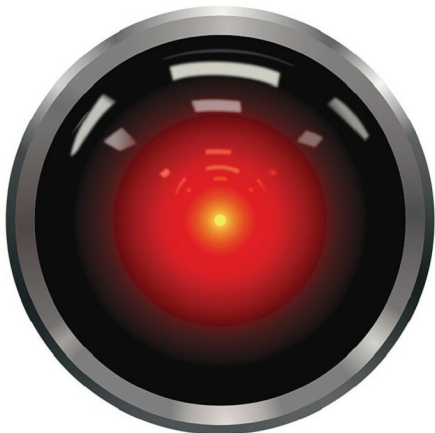
be entirely clear to AI researchers, the neural network sometimes can yield odd, counterintuitive results like the Twitter bot we just talked about.

"A study of visual classification AIs using neural networks found that while AIs were able to generally identify objects as well as humans, in some cases AIs made confident identifications of objects that were not only incorrect but that looked vastly different from the purported object to human eyes. The AIs interpreted images, which to the human eye looked like static or abstract wavy lines, as animals or other objects, and asserted greater than 99.6 per cent confidence in their estimation," explains Paul Scharre, senior fellow and director of 20YY Future of Warfare Initiative, in *Autonomous Weapons and Operational Risk*.

Deep neural networks and Narrow AI

Existing technology for Tesla's semi-autonomous driving cars uses deep neural networks (DNNs). When the human brain tries to recognise objects in images, it does not see pixels; instead it sees edges on them. A DNN tries to recreate how a human brain functions by programming it to only recognise edges.

This machine or program uses code to look at objects and do some unsupervised learning of its own—known as machine learning. When enough time has passed, the machine learns to distinguish between whatever the DNN was told to look out for. Intelligence of the DNN de-



The popular camera eye of HAL 9000, an artificial intelligence from the movie '2001: A Space Odyssey'

pends on its processing power and the time spent learning.

Companies like Tesla are developing Narrow AI, which is a variant that stands for a non-sentient computer intelligence that is focused on a narrow task that prevents it from doing something like trying to take over the world. Apple's Siri is apparently a good example of Narrow AI.

Specialised hardware to run neural networks

Neural networks used for image analysis are typically run on graph-

ics processing units that specialise in image processing. But much more can be done to make circuits that run neural networks efficiently.

Joel Emer, Massachusetts Institute of Technology (MIT) computer science professor and senior distinguished research scientist at NVIDIA, has developed Eyeriss, a custom chip designed to run a state-of-the-art convolutional neural network.

In February 2016, IEEE International Solid-State Circuits Conference in San Francisco, USA, teams from MIT, NVIDIA and Korea Advanced Institute of Science and Technology (KAIST) showed prototypes of chips designed to run artificial neural networks. The KAIST design minimises data movement by bringing memory and processing closer together. It saves energy by both limiting data movement and by lightening the computational load. Kim's group observed that

Are our existing automated defence systems autonomous

Weapons bring in the idea of offence as well as attack. However, specialised weapons for defence keep cropping up from time to time. Like Patriot, autonomous defensive weapons (Skyнет and Arcnet are two fictional examples) have made their presence known and very soon we could see some offensive ones, too. However, "Defensive systems are under scrutiny to be considered autonomous weapons or not," says Dr Peter Asaro, co-founder and vice-chair, International Committee for Robot Arms Control. Moreover, automated defence systems are not sentient (able to perceive or feel), yet.